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DEPARTMENT OF DEFENSE
INTERFACE STANDARD

LAMPS, REFLECTORS AND ASSOCIATED SIGNALING EQUIPMENT FOR MILITARY VEHICLES

AMSC N/A FSC 2310, 2320, 2330, 2340, 2350

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Lamps, Reflectors and Associated Signaling Equipment for Military Vehicles

MIL-STD-1179D

1. This Military Standard is approved for use by U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

This standard establishes the requirements for lamps and reflectors to enable safe operation of military vehicles in darkness and other conditions of reduced visibility, and during blackout conditions. The standard also establishes the requirements for associated signaling equipment to enable safe operation of military vehicles under normal conditions of visibility as well as in darkness and other conditions of reduced visibility.

The requirements of this standard have been established in consideration of the following, listed in order of precedence:

a. Military mission requirements.
b. Motor Vehicle Safety Standard (MVSS) No. 108.
c. Greater component damage potential due to the type of usage on military vehicles.
d. Requirements of other countries.
e. Existing practice on military vehicles.

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1. SCOPE

1.1 Scope. This standard applies to all military combat, special purpose and tactical transport vehicles, both tracked and wheeled. Administrative and commercial vehicles are not within the scope of this standard, since total compliance with Motor Vehicle Safety Standard (MVSS) No. 108 is required for these vehicles.

1.2 Purpose. The purpose of this standard is to establish uniform requirements for lamps, reflectors and associated signaling equipment for new military vehicles, tactical, combat and special purpose.

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

STANDARDS
MILITARY

MS18003	- Lamp, Incandescent - Sealed Beam, Par 36, Headlamp, 3 Contact Lug Type.
MS18008	- Lamp, Incandescent - Sealed Beam, Par 56, Headlamp, 3 Cable Type.
MS35387	- Reflector, Indicating, Clearance.
MS35423	- Light, Marker, Clearance - Service.
MS51113	- Switch, Vehicular Lights: 24 Volt DC (Waterproof).
MS51317	- Light, Warning, Vehicular, Rotating, DC.
MS51318	- Headlight: Blackout, 24 Volt, Waterproof.
MS51329	- Stop Light-Tail Light, Vehicular - 24 Volt, Service Stop, Service Tail, Blackout Tail.
MS51330	- Stop Light-Tail Light, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS52125	- Composite Light-Tail, Stop, Turn and Marker.
MS52126	- Composite Light-Front Turn, Park and Marker.
MS53022	- Light, Composite, Vehicular, Blackout Driving, Blackout Marker, Service Infrared - 24 Volt.
MS75064	- Switch, Rotary: Stoplight, Vehicular, Mechanically Actuated, 24 Volt DC.

DRAWINGS
MILITARY

7972325	- Headlamp Assembly.
10944453	- Lamp Assembly, Sealed Unit, Signal.
10944454	- Lamp Assembly, Sealed Beam Unit, Spot Lamp.
11613631	- Flasher, Turn Signal.
11613632	- Control Assembly, Directional (Waterproof).
11614156	- Composite Light - Front Turn, Park and Marker.
11614157	- Composite Light - Tail, Stop, Turn and Marker.
11614174	- Reflector, Tape/Adhesive Backing.
11668932	- Lamp Assembly.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific acquisition functions should be obtained from the acquisition activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
SAE Standards and Recommended Practices

J222	- Parking Lamps (Front Position Lamps).
J564a	- Headlight Beam Switching.
J565b	- Semiautomatic Headlamp Beam Switching Devices.
J566 Jan 1966	- Headlamp Mountings.
J571d	- Dimensional Specifications for Sealed Beam Headlamp Units.
J575	- Tests for Motor Vehicle Lighting Devices and Components.
J578c	- Color Specification for Electric Signal Lighting Devices.
J579a, c	- Sealed Beam Headlamp Units for Motor Vehicles.
J580a, b	- Sealed Beam Headlamp Assembly.
J585e	- Tail Lamps (Rear Position Lamps).
J586c	- Stop Lamps for use on Motor Vehicles Less Than 2032 mm in Overall Width.
J588e	- Turn Signal Lamps.
J589 Apr 1964	- Turn Signal Switch.
J590b	- Turn Signal Flasher.
J592e	- Clearance, Side Marker, and Identification Lamps.
J593c	- Backup Lamps (Reversing Lamps).
J594f	- Reflex Reflectors.
J910 Jan 1966	- Hazard Warning Signal Switch.
J945 Feb 1966	- Vehicular Hazard Warning Signal Flasher.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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DEPARTMENT OF TRANSPORTATION (DOT)

Motor Vehicle Safety Standard (MVSS) No. 108

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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ry of the for operation of four electric switches mounted on the periphe

3. DEFINITIONS

3.1 Headlamps. A headlamp including one or more sealed beam units is a major lighting device used to provide general illumination ahead of the vehicle.

3.2 Taillamps. Taillamps are those lamps used to designate the rear of a vehicle by a steady warning light other than clearance or identification lamps.

3.3 Stoplamps. Stoplamps are those lamps giving a steady warning light to the rear of a vehicle or train of vehicles, to indicate the application of the service brake pedal and thus the intention of the operator of a vehicle to diminish speed or stop.

3.4 Turn signal lamps. Turn signal lamps are those which indicate a change in direction of a vehicle by giving flashing warning lights on the side toward which the turn will be made.

3.5 Turn signal control unit. The control unit is that part of a turn signal system by which the operator of a vehicle causes the signal units to function.

3.6 Turn signal flasher. The turn signal flasher is a device which causes a turn signal lamp to be operated intermittently as long as it is turned on.

3.7 Clearance lamps. Clearance lamps are lamps located on the left and right at the front and rear of a vehicle. The front ones show to the front and outboard side, the rear ones show to the rear and outboard side indicating the overall width of the vehicle.

3.8 Side-marker lamps. Side-marker lamps are lamps located on the left and right sides near the front and rear of a vehicle which show to the side, to mark the length of the vehicle.

3.9 Identification lamps. Identification lamps are lamps used in groups of three in a horizontal row centered on the vehicle at the front or rear, or on both the front and rear to identify certain types of vehicles.

3.10 Reflex reflectors. Reflex reflectors are devices which are used on a vehicle to alert an approaching driver, by reflected light from the head lamps, of the presence of another vehicle/object.

3.11 Vehicular hazard warning flasher. A vehicular hazard warning flasher is a device which, as long as it is turned on, causes all turn signal lamps to flash.

3.12 Vehicular hazard warning flasher switch. A vehicle hazard warning switch is a driver controlled device which causes all turn signal lamps to flash simultaneously to indicate to approaching drivers the presence of vehicular hazard.

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3.13 Intermediate side-marker lamps. Intermediate side-marker lamps are those lamps on vehicles which are located on, or in proximity to, the midpoint of the vehicle. (Intermediate side-marker lamps are required when the overall vehicle length is 30 feet or more.)

3.14 Intermediate side-reflex reflector. Intermediate side-reflex reflectors are those reflectors on vehicles which are located on, or in proximity to, the midpoint of the vehicle. (Intermediate side-reflex reflectors are required when the overall vehicle length is 30 feet or more.)

3.15 Parking lamps. Parking lamps are multi-purpose lights facing forward and located on the front of the vehicle, one on each side and as far apart as practicable. They serve as clearance lamps indicating the width of the vehicle, as emergency head lamps, as low power illumination for parked vehicles and include turn signal capability.

3.16 Light, warning, vehicular, rotating, DC. A 360 degree flashing warning signal lamp for use on emergency vehicles, repair vehicles, utility servicing vehicles, truck tractors designed to haul oversize slow moving loads, truck wreckers, and on other vehicles that frequently deviate from or obstruct normal traffic patterns. Also used as Convoy Warning Lights (CWL).

3.17 Flashing warning lamps. Flashing warning lamps are lamps mounted on vehicles to give a flashing warning to traffic.

3.18 Back-up lamp. A back-up lamp is a lamp on the rear of a vehicle to indicate that the vehicle is in reverse gear and the intention of the operator is to back up the vehicle.

3.19 Blackout lamps. Blackout lamps are lamps designed to maintain vehicle blackout security during blackout conditions by limiting horizontal and vertical illumination as well as providing ample light for safe vehicle operation.

3.20 Blackout marker lamps. Blackout marker lamps are lamps used on military vehicles to indicate its width. Under MVSS terminology they are referred to as clearance lamps.

3.21 Military wheeled vehicle overall width. Overall width for military wheeled vehicles is determined with doors and windows closed, and the wheels in the straight-ahead position. The term "overall width" refers to the nominal design dimension of the widest part of the vehicle, exclusive of signal lamps, marker lamps, outside rear view mirrors, flexible fender extensions, and mud flaps.

3.22 Convoy Warning Light (CWL). See 3.16.

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4. GENERAL REQUIREMENTS

4.1 General. Requirements that are common to both tactical (wheeled) and combat (tracked) vehicles for service lighting equipment.

4.1.1 Lamps, tail, stop, and turn candlepower. A parking lamp, tail lamp, stop lamp, or turn signal lamp shall meet the minimum percentage specified in figure 1a of the corresponding minimum allowable value specified in figure 1b. The maximum candlepower output of each stop, turn signal, tail and parking lamp shall not exceed that prescribed in figure 1b. The values specified in figures 1a and 1b are substituted for those specified in table I of the following SAE Standards: J222, Parking Lamps; J585e, Tail Lamps (at H or above); J586c, Stop Lamps and J588e, Turn Signal Lamps. Figures 1a and 1b are shown below.

Test points (deg)		Turn signal	Stop	Parking	Tail
10U, 10D....	5L, 5R.....	20	20	20	20
5U, 5D.....	20L, 20R.....	12.5	12.5	10	15
	10L, 10R.....	37.5	37.5	20	40
	V.....	87.5	87.5	70	90
H.....	10L, 10R.....	50	50	35	40
	5L, 5R.....	100	100	90	100
	V.....	100	100	100	100

Note: Minimum design candlepower requirements are determined by multiplying the percentages given in the figure by the minimum allowable candlepower values in figure 1b. The resulting values shall be truncated after one digit to the right of the decimal point.

FIGURE 1a. Required percentages of minimum candlepower of figure 1b.

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Lamp	Lighted sections		
	1	2	3
Stop [1]	80/300	95/360	110/420
Tail [2]	2/18	3.5/20	5.0/25
Parking [2]	4.0/125	.....	.....
Red turn signal.....	80/300	95/360	110/420
Yellow turn signal rear.....	130/750	150/900	175/1050
Yellow turn signal front [3] ..	200/-	240/-	275/-
Yellow turn signal front [3] ...	500/-	600/-	685/-

1 Maximum at H or above.
2 The maximum candlepower value of 125 applies to all test points at H or above. The maximum allowable candlepower value below H is 250.
3 Values apply when the optical axis (filament center) of the front-turn signal is at a spacing less than 4 in (10 cm) from the lighted edge of the headlamp unit providing the lower beam, or from the lighted edge of any additional lamp installed as original equipment or used in lieu of the lower beam.

FIGURE 1b. Minimum and maximum allowable candlepower values.

4.1.2 Lamp candlepower. A parking lamp, taillamp, stoplamp or turn signal lamp is not required to meet the minimum candlepower value at each test point specified in this standard if the sum of the percentages of the minimum candlepower measured at the test points is not less than that specified for each group listed in figure 1c (see below).

Group and test points	Turn signal	Stop	Parking	Tail
1 10U-5L, 5U-20L, 5D-20L, 10D-5L.....	65	65	60	70
2 5U-10L, H-10L, 5D-10L....	125	125	75	120
3 H-5L, 5U-V, H-V, 5D-V, H-5R.....	475	475	420	480
4 5U-10R, H-10R, 5D-10R....	125	125	75	120
5 10U-5R, 5U-20R, 5D-20R, 10D-5R.....	65	65	60	70

FIGURE 1c. Sum of the percentages of grouped minimum candlepower.

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primary power is available. The erection system shall be driven between

4.1.3 Intermediate turn signal lamp visibility. Intermediate turn signal lamps shall be visible through an angle of 60 degrees (minimum) outward from the longitudinal axis of a vehicle. A blind angle of 5 degrees outward from the edge of the vehicle is permissible.

4.1.4 Lamp color intensity. The color in all lighting equipment covered by this standard shall comply with SAE J578c, and shall meet the minimum candlepower requirements specified in 4.1.2.

4.2 Vehicle lighting equipment location.

4.2.1 Headlamp location. Headlamps shall be located on the front of the vehicle and as shown in tables II, IV and V so that they will clear all portions of the vehicle and its appurtenances.

4.2.2 Lamp and reflector location. Lamps and reflectors shall be located so that their beam pattern and visibility shall be unobstructed by any part of the vehicle, throughout the range of the photometric and visibility angles specified. Lamps and reflectors shall not be mounted on extension brackets or in any other manner that will increase their damage potential beyond that of lamps and reflectors mounted directly on the vehicle.

4.2.3 Service lighting equipment location. When lamps, reflective devices, and associated signaling equipment are required to be "within X inches of the extreme edge of the vehicle" or "within X inches of the extreme front (rear) of the vehicle", measurements shall be taken as follows:

a. Measurements from the extreme edge of the vehicle shall be from a vertical plane tangent to the outer edge of the vehicle (exclusive of signal lamps, marker lamps, outside rear view mirrors, with doors and windows closed and wheels in a straight-ahead position) and parallel to the longitudinal axis of the vehicle, to the outboard edge of the luminous surface of the lamp or reflector, or where a multiple-compartment lamp or multiple lamps are used, to the outboard edge of the luminous surface of the outermost compartment or lamp.
b. Measurements from the extreme front (rear) edge of the vehicle shall be from a vertical plane tangent to the extreme front (or rear) edge of the vehicle (exclusive of winch projections or trailer tongues) and perpendicular to the longitudinal axis of the vehicle, to the front (or rear) edge of the luminous surface of the lamp or reflector, or where a multiple compartment lamp or multiple lamps are used, to the front (or rear) edge of the luminous surface of the forward (or rearward) compartment or lamp.

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4.3 Lamp and equipment combinations.

4.3.1 Lamp, reflector, or associated equipment combination. Two or more lamps, reflective devices or items of associated equipment may be combined if the requirements for each lamp, reflective device, and item of associated equipment are met, except that no clearance lamp may be combined optically with any tail lamp or identification lamp, and no high-mounted stop lamp shall be combined with any other lamp or reflective device.

4.3.1.1 Turn signal lamp combination. No turn signal lamp may be combined optically with any lamp (other than a stop lamp) that produces more than 1/5 the light intensity of the turn signal lamp at test points of H-V, H-5L, H-5R, and 5U-V, nor more than 1/3 the intensity of any test point on or above the horizontal. (See SAE J575 for definition of test points.)

4.3.1.2 Turn signal and stoplamp combinations. No turn signal lamp may be combined optically with a stop lamp unless the stop lamp is extinguished when the turn signal is flashed.

4.3.2 Headlamp combination, candlepower. When a parking lamp is optically combined with a turn signal, and the parking lamp is connected to be operated with the headlamps, the turn signal shall not be less than three times the candlepower of the parking lamp at any test point on or above horizontal, except that at H-V, H-5L, H-5R, and 5U-V, the turn signal shall not be less than five times the candlepower of the parking lamp. (See SAE J575 for definition of test points.)

4.4 Service lamp wiring requirements.

4.4.1 Stoplamp actuation. Stoplamps shall be actuated upon application of service brakes provided the vehicular light switch is in stop light, service drive or park position.

4.4.2 Hazard warning signal actuation. The vehicular hazard warning signal operating unit shall operate independently of the ignition or equivalent switch, and when energized shall cause all turn signal lamps to flash simultaneously, provided the vehicular light switch is in stop light, service drive, or park position.

4.4.3 Head, parking and clearance lamp actuation. On all vehicles, parking lamps, clearance lamps, side-marker lamps and taillamps shall be energized when the headlamps are energized. The taillamps, identification lamps (if applicable) and side-marker lamps shall be energized when either the parking or clearance lamps are energized.

4.4.4 Headlamp, upper and lower beam switch. Each vehicle shall be equipped with a means of switching between upper and lower headlamp beams, conforming to SAE J564a or SAE J565b. Each vehicle shall be equipped with a means of indicating to the driver when the upper beams are on, conforming to SAE J564a, except for signal color.

4.5 General blackout lighting requirements. No service lamp shall be capable of being activated while the vehicle's light switch (MS51113) is in blackout marker, blackout drive or off position.

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4.5.1 Vehicle blackout lights and location. The minimum number of blackout lights on a military vehicle shall be as specified in table VI.

4.5.2 Blackout marker lamps. As referenced in table VI, the blackout marker lamps shall meet the requirements specified in 4.5.2.1 through 4.5.2.4.

4.5.2.1 Visibility of beams. The front position marker filter lens shall be opaque except for two Y-shaped openings which shall emit white light. The two openings of each blackout marker lamp shall be individually visible up to 60 feet. Beyond 60 feet the opening shall appear as a single point of light. The opening shall be visible at a distance of 800 feet and shall not be visible beyond 1600 feet.

4.5.2.2 Visibility from air.

a. Vehicle on 20 percent upgrade. When vehicle is on a 20 percent upgrade, the vertical angles of cutoff above the horizontal shall be the maximum practicable provided the single light effect of the front marker shall not be visible from the air at more than 400 feet above a horizontal line extended through the blackout marker lamp.
b. Vehicle on level road. When vehicle is on a level road, the vertical angles of cutoff of the marker below the horizontal shall be such as not to cast a reflection on the road that would be visible from the air at a height of more than 400 feet.

4.5.2.3 Horizontal visibility on road. When the vehicle is on a level road, the horizontal angles of cutoff shall be the maximum practicable, but shall be not less than 60 degrees right or left of the beam centerline at 100 feet.

4.5.2.4 Color of light. Color values of the transmitted light shall be in accordance with trichromatic coefficients specified in component drawings.

4.5.3 Blackout driving lamp. The blackout driving lamp, at a design voltage of 28.0 volts (V) direct current (dc), shall emit a white light that has a candlepower of 25 minimum and 50 maximum measured in the most intense portion of the beam 10 feet from its source. Its intensity shall decrease uniformly 20 to 100 feet from the light with the top of the light beam directed not less than two degrees from horizontal. The beam distribution on a level road at a point 100 feet from the vehicle blackout driving light shall be 30 feet wide.

4.5.4 Blackout taillamp. The blackout taillamp shall meet the requirements specified in 4.5.4.1 through 4.5.4.4.

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4.5.4.1 Visibility of beams. The blackout taillamp filter lens shall be opaque except for four Y-shaped openings which shall emit red light. The four openings of each blackout taillamp shall be individually visible up to 60 feet. From 60 to 120 feet, the openings shall appear as two points of light. The lighted openings shall be visible at a distance of 800 feet and shall not be visible beyond 1600 feet.

4.5.4.2 Visibility from air.

a. Vehicle on 20 percent downgrade. When vehicle is on a 20 percent downgrade, the vertical angles of cutoff above the horizontal shall be the maximum practicable, provided the single light effect of the blackout taillamp shall not be visible from the air at more than 400 feet above a horizontal line extended through the blackout taillamp.
b. Vehicle on level road. When vehicle is on a level road the vertical angles of cutoff of the blackout taillamp below the horizontal shall be such as not to cast a reflection on the road that would be visible from the air at a height of not more than 400 feet.

4.5.4.3 Horizontal visibility on road. When the vehicle is on a level road, the horizontal angles of cutoff shall be the maximum practicable but shall be not less than 60 degrees right or left of the beam centerline at 100 feet.

4.5.4.4 Color of light. Color values of the transmitted light shall be in accordance with trichromatic coefficients specified in component drawings.

4.5.5 Blackout stoplamp. The blackout stoplamp shall meet the requirements specified in 4.5.5.1 through 4.5.5.4.

4.5.5.1 Visibility of beam. The blackout stoplamp filter lens shall be opaque except for the Y-shaped opening which shall emit amber light. The lighted opening shall be visible at a distance of 800 feet and shall not be visible beyond 1600 feet.

4.5.5.2 Visibility from air.

a. Vehicle on 20 percent downgrade. When vehicle is on a 20 percent downgrade, the vertical angles of cutoff above the horizontal shall be the maximum practicable, provided the single light effect of the blackout stoplamp shall not be visible from the air at more than 400 feet above a horizontal line extended through the blackout taillamp.
b. Vehicle on level road. When vehicle is on a level road the vertical angles of cutoff of the blackout stoplamp below the horizontal shall be such as not to cast a reflection on the road that would be visible from the air at a height of more than 400 feet.

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4.5.5.3 Horizontal visibility on road. When the vehicle is on a level road, the horizontal angles of cutoff shall be the maximum practicable but shall be not less than 60 degrees right or left of the beam centerline at 100 feet.

4.5.5.4 Color of light. Color values of the transmitted light shall be in accordance with trichromatic coefficients specified in component drawings.

4.6 Wheeled vehicle requirements for service lighting equipment.

4.6.1 Wheeled vehicle minimum lighting equipment. Except as specified in 4.6.1.1 through 4.6.1.7, military wheeled vehicles shall be equipped as a minimum with lamps, reflective devices, and associated signaling equipment in accordance with table I or table III, as applicable. SAE Standards and Recommended Practices subreferenced by the SAE Standards and Recommended Practices included in tables I and III and Paragraphs S4.1.4 and S4.5.1 are those published in the 1970 edition of the SAE Handbook, except that the SAE standard referred to as "J575" is J575e, Tests for Motor Vehicle Lighting Devices and Components, August 1970, for stoplamps, taillamps, and turn signal lamps designed to conform to SAE Standards J586c, J585d/J585e, and J588, respectively, and for highmounted stoplamps designed to conform to SAE Recommended Practice J186a. The subreferenced Standards and Recommended Practices for headlamps designed to conform to SAE Standards J579c, Sealed Beam Headlamp Units, December 1974, are those published in the 1977 edition of the SAE Handbook.

4.6.1.1 Wheeled vehicle identification lamps. Front identification lamps shall be applied only on hardtop vehicles.

4.6.1.2 Truck and truck tractor tail and parking lamps. When the tail-lamps and parking lamps on trucks and truck tractors are located within 6 inches of the extreme edge of the vehicle (see 4.2.2), they may also serve as clearance lamps, and separate clearance lamps are not required.

4.6.1.3 Truck tractor side marker devices, clearance lamps and identification lamps. A truck tractor need not be equipped with any rear side marker devices, rear clearance lamps and rear identification lamps.

4.6.1.4 Wheeled vehicle intermediate marker lamp and side reflector. Intermediate side marker lamps and intermediate side reflex reflectors are required only on vehicles 30 or more feet in overall length.

4.6.1.5 Truck, light, warning, vehicular, rotating DC. The 360 degree flashing warning lamp is required only on trucks designed specifically to haul overwidth, slow-moving loads and on wreckers, emergency vehicles, repair vehicles, and utility servicing vehicles.

4.6.1.6 Short trailer side marker lamps and front side reflector. Trailers less than 6 feet in overall length, including the trailer tongue, need not be equipped with front side marker lamps and front side reflex reflectors.

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4.6.1.7 Narrow trailer reflector and lamp. A trailer that is less than 80 inches in overall width may be equipped with only one of each of the following reflective devices, located at or near its vertical centerline: taillamp, stoplamp, and rear reflex reflector.

4.6.1.8 Semitrailer convertor dolly reflector and lamp. No lighting devices or reflectors are required on any semitrailer converter dolly.

4.7 Wheeled vehicle service lighting equipment location.

4.7.1 Truck and trailer lamp and reflector location. Except as specified in 4.7.1.1 through 4.7.1.2, lamps, reflective devices, and associated equipment shall be located in accordance with table II or table IV, which ever applies.

4.7.1.1 Truck tractor, rear red reflex reflector location. On truck tractors, the red reflex reflector may be mounted on the back of the cab, at a height not less than 4 inches above the rear tires and must indicate the overall width of the vehicle (see 4.2.2).

4.7.1.2 Wheeled vehicle rear identification lamp location. When the rear identification lamps are mounted at the extreme height of a vehicle, rear clearance lamps need not be located as close as practicable to the top of the vehicle.

4.8 Wheeled vehicle service lamp wiring requirements.

4.8.1 Back-up lamp actuation. Back-up lamps shall be actuated when the vehicle gear shift lever is placed in reverse gear, provided the vehicular light switch is in stop light, service drive or park position.

4.9 Tracked vehicle service lighting equipment and location.

4.9.1 Minimum lighting equipment. Tracked vehicles shall be equipped, at minimum with lamps and reflective devices as specified in table V.

4.10 International standardization agreements. Certain provisions of this Military Standard are the subject of international standardization agreements; CENTO SANAG 2024 (Second Edition) and NATO STANAG 2024.

4.10.1 Clearance lights. All tracked and tactical wheeled vehicles, regardless of width, must have two (one on each side as far outboard as practicable) front mounted clearance lights, white or amber in color, if the headlights are not within 16 inches of the edge of the vehicle.

4.10.2 Reflex reflectors. Reflex reflectors must be attached as close as practicable to the outer edge of each side, and at the same height (see 4.2.2).

4.10.3 Provisions for tracked vehicle CWL. Provide a fused weatherproof power receptacle on the top of the vehicle (or turret) provide power for a CWL. The circuitry shall be such that there will be no power to the receptacle when the vehicle is in the blackout mode. The CWL should be readily attachable and removable.

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4.10.4 Provisions for wheeled vehicle CWL. Provide two waterproof receptacles on the top of the vehicle to provide power for a CWL. A separately fused, waterproof on/off switch shall be located in the cab within easy reach of the driver. The circuitry shall be such that there will be no power to the receptacle when the vehicle is in the blackout mode.

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5. DETAILED REQUIREMENTS. Not applicable.

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TABLE I. Lighting equipment requirements for tactical military trucks and trailers of less than 80 inches overall width (see 3.21).

Items	Quantity and color 1/		Applicable SAE standards and military part numbers
	Trucks	Trailers	
Headlamps	2 white, 7 inch		J580a, b MS18008-4863
Taillamps	2 red	2 red	J585e 11614157
Stoplamps	2 red	2 red (see 4.1.1)	J586c 11614157
Parking lamps	2 amber (see 4.1.1)		J222 11614156
Reflex reflectors	4 red; 2 amber	4 red; 2 amber (see 4.6.1.7 & 4.6.1.8)	J594f MS35387-1 or 11614174-1 red MS35387-2 or 11614174-2 amber
Intermediate side-reflex reflectors	2 amber (see 4.6.1.3)	2 amber (see 4.6.1.3)	J594f MS35387-2 or 11614174-2
Intermediate side-marker lamps	2 amber (see 4.6.1.3)	2 amber (see 4.6.1.3)	J592e MS35423-1
Side marker lamps	2 red; 2 amber	2 red; 2 amber (see 4.6.1.6)	J592e MS35423-1 amber MS35423-2 red
Back-up lamp	1 white		J593c
Turn signal lamps	2 red 2 amber	2 red 2 amber	J588e 11614156 amber
Turn signal control unit	1		J589 Apr 1964 11613632
Turn signal flasher	1		J590b 11613631

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TABLE I. Lighting equipment requirements for tactical military trucks and trailers of less than 80 inches overall width (see 3.21) - Continued.

Items	Quantity and color 1/		Applicable SAE standards and military part numbers
	Trucks	Trailers	
Vehicular hazard warning signal operating unit	1		J910 Jan 1966 11613632
Vehicular hazard warning signal flasher	1		J945 Feb 1966 11613631
Light, warning, vehicular, rotating, DC	1		MS51317
Receptacles for CWL	2		
Clearance lamps	2 red; 2 amber	2 red; 2 amber	J592e MS35423-1 amber MS35423-2 red

1/ Color in accordance with SAE J578c.

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TABLE II. Lighting equipment location for tactical military trucks and trailers of less than 80 inches overall width (see 3.21).

Equipment	Location on trucks and trailers	Height above road surface measured from center of item on vehicle at curb weight
Head lamps	On the front, the same height, 1 on each side of the vertical centerlineas far apart as practical. Not required on trailers.	Not less than 22 inches, nor more than 54 inches.
Tail lamps	On the rear - 1 on each side of the vertical centerline, at the same level, and as far apart as practical (see 4.6.1.2).	Not less than 15 inches, nor more than 72 inches.
Stop lamps	On the rear - 1 on each side of the vertical centerline, at the same height, and as far apart as practical.	Not less than 15 inches, nor more than 72 inches.
Parking lamps	On the front - 1 on each side of the vertical centerline, at the same level, and as far apart as practical. Not required on trailers (see 4.6.1.2).	Not less than 15 inches, nor more than 72 inches.
Reflex reflectors	On the rear - 1 red on each side of the vertical centerline, at the same level, and as far apart as practical. On each side - 1 red as far to the rear as practicable, and 1 amber as far to the front as practical (see 4.6.1.6).	Not less than 15 inches, nor more than 60 inches.
Back-up lamp	On the rear.	No requirement.
Turn signal lamps	At or near the front - 1 amber on each side of the vertical centerline, at the same height, and as far apart as practical. Not required on trailers. On the rear - 1 red or amber on each side of the vertical centerline, at the same height and as far apart as practical.	On the front - not less than 15 inches, nor more than 83 inches. On the rear - not less than 15 inches, nor more than 83 inches.
Side marker lamps	On each side - 1 red as far to the rear as practical, and 1 amber as far to the front as practical (see 4.6.1.6) and be visible from a point in front of the vehicle.	Not less than 15 inches.

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TABLE II. Lighting equipment location for tactical military trucks and trailers of less than 80 inches overall width (see 3.21) - Continued.

Equipment	Location on trucks and trailers	Height above road surface measured from center of item on vehicle at curb weight
Intermediate side marker lamps	On each side - 1 amber located at or near the mid-point between the front and rear side marker lamps (see 4.6.1.3).	Not less than 15 inches.
Intermediate reflex	On each side - 1 amber located at or near the mid-point between the front and rear marker reflectors (see 4.6.1.4).	Not less than 15 inches, nor more than 60 inches.
Clearance lamps (If headlamps are located more than 16 inches from edge of vehicle.)	On the front - 1 amber or white on each side of the vertical centerline, at the same level, as far apart as practical.	Not less than 15 inches.
Light, warning, vehicular, rotating, DC	On top - near the centerline of the vehicle.	
Receptacle for CWL	Two waterproof receptacles located on the exterior of the back of the cab, one on each side of the center line of the vehicle as far outboard as practicable, where the CWL may be plugged in. Not required on trailers.	

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TABLE III. Lighting equipment requirements for tactical military trucks, trailers and truck tractors of 80 or more inches overall width (see 3.21).

Items	Quantity and color 1/			Applicable SAE standards and military part numbers
	Trucks	Trailers	Truck-tractors	
Headlamps	2 white, 7-inch		Same as trucks	J580a, b MS18008-4863
Taillamps	2 red	Same as trucks	Same as trucks	J585e 11614157
Stoplamps	2 red	Same as trucks	Same as trucks	J586c 11614157
Reflex reflectors	4 red; 2 amber	4 red; 2 amber	2 red; 2 amber	J594f MS35387-1 or 11614174-1 red MS35387-2 or 11614174-2 amber
Side marker lamps	4 red; 2 amber	4 red; 2 amber (see 4.6.1.6)	2 amber	J592e MS35423-1 amber MS35423-2 red
Back-up lamp	1 white		1 white	J593c
Turn signal lamps	2 red or 2 amber (rear) 2 amber (front)	2 red	Same as trucks	J588e 11614156 amber 11614157 red
Turn signal control unit	1		1	J589 Apr 1964 11613632
Turn signal flasher	1		1	J590b 11613631
Vehicular hazard warning signal operating unit	1		1	J910 Jan 1966 11613632
Vehicular hazard warning signal flasher	1		1	J945 Feb 1966 11613631

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TABLE III. Lighting equipment requirements for tactical military trucks, trailers and truck tractors over 80 inches overall width (see 3.21) - Continued.

Items	Quantity and color 1/			Applicable SAE standards and military part numbers
	Trucks	Trailers	Truck-tractors	
Identification lamps	3 red; 3 amber (see 4.6.1.1)	3 red (see 4.6.1.1)	3 amber, 3 red (see 4.6.1.1)	J592e MS35423-1 amber MS35423-2 red
Clearance lamps	2 red; 2 amber (see 4.3.1 & 4.6.1.2)	2 red; 2 amber	2 amber, 2 red (see 4.3.1 & 4.6.1.2)	J592e MS35423-1 amber MS35423-2 red
Parking lamps	2 amber		Same as trucks	J222 11614156
Intermediate side marker lamps	2 amber (see 4.6.1.3)	2 amber (see 4.6.1.3)	Same as trucks	J592e MS35423-1
Intermediate side reflex reflectors	2 amber (see 4.6.1.3)	2 amber (see 4.6.1.3)	Same as trucks	J594f MS35387-2 or 11614174-2
Light, warning, vehicular, rotating, DC	1 lamp with color as specified by local ordnance (see 4.6.1.5)		1 amber (see 4.6.1.5)	MS51317
Receptacle for CWL	2		2	

1/ Color in accordance with SAE J578c

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TABLE IV. Lighting equipment location for tactical military trucks, trailers and truck tractors of 80 or more inches overall width (see 3.21).

Item	Location on			Height above road surface measured from center of item on vehicle at curb weight
	Trucks	Trailers	Truck-tractors	
Head lamps	On the front, at the same height, 1 on each side of the vertical centerline; as far apart as practicable.		Same as trucks	Not less than 22 inches, nor more than 54 inches.
Tail lamps	On the rear - 1 on each side of the vertical centerline at the same level, as far apart as practicable (see 4.6.1.2).	Same as trucks	Same as trucks	Not less than 15 inches, nor more than 72 inches.
Stop lamps	On the rear - 1 on each side of the vertical centerline at the same level, and as far apart as practicable.	Same as trucks	Same as trucks	Not less than 15 inches, nor more than 72 inches.
Reflex reflectors	On the rear - 1 red on each side of the vertical centerline, at the same level, as far apart as practicable.	Same as trucks	Same as trucks	Not less than 15 inches, nor more than 60 inches.
	On each side - 1 red as far to the rear as practicable, and 1 amber as far to the front as practicable.	Same as trucks	On each side - 1 amber as far to the front as practicable.	Not less than 15 inches, nor more than 60 inches.

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TABLE IV. Lighting equipment location for tactical military trucks, trailers and truck tractors of 80 or more inches overall width (see 3.21) - Continued.

Item	Location on			Height above road surface measured from center of item on vehicle at curb weight
	Trucks	Trailers	Truck-tractors	
Side marker lamps	On each side - 1 red as far to the rear as practicable, and 1 amber as far to the front as practicable.	Not less than 15 inches and on the rear of trailers not more than 60 in (see 4.6.1.6).	On each side - 1 amber as far to the front as practicable.	Not less than 15 inches.
Back-up lamps	On the rear.			No requirement.
Turn signal lamps	At or near the front 1 amber on each side of the vehicle centerline, at the same level, as far apart as practicable.	Same as trucks	Same as trucks	On the front - not less than 15 inches, nor more than 83 inches.
	On the rear, 1 red or amber on each side of the vertical centerline at the same height and as far apart as practicable. The turn signal may be part of the stoplight assembly.	Same as trucks	Same as trucks	On the rear - not less than 15 inches, nor more than 83 inches.

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TABLE IV. Lighting equipment location for tactical military trucks, trailers and truck tractors of 80 or more inches overall width (see 3.21) - Continued.

Item	Location on			Height above road surface measured from center of item on vehicle at curb weight
	Trucks	Trailers	Truck-tractors	
Identification lamps	On the front, 3 amber lamps and on the rear, 3 red lamps. Grouped in a horizontal row with lamp centers spaced not less than 6 inches nor more than 12 inches apart, and mounted as close as practicable to the vertical centerline and the top of the vehicle (see 4.6.1.1 and 4.7.1.2).	On the rear, 3 red lamps grouped same as for trucks (see 4.6.1.1).	On the front, 3 amber lamps grouped same as for trucks (see 4.6.1.1).	On the front only. No part of the clamp or mountings shall extend below the vehicle windshield.
Clearance lamps	On the front and rear - 2 amber or white lamps on front, 2 red on rear to indicate overall width of the vehicle, one on each side of vertical centerline as far apart as practical.	On the front and on the rear - 1 on each side of the vertical centerline.	On the front - 1 on each side of the vertical centerline as far apart as practical (see 4.6.1.3 and 4.10.1).	No requirement.
Parking lamps	On the front - 1 on each side of the vertical centerline, at the same level (see 4.6.1.2).		Same as trucks	Not less than 15 inches, nor more than 72 inches.

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TABLE IV. Lighting equipment location for tactical military trucks, trailers and truck tractors of 80 or more inches overall width (see 3.21) - Continued.

Item	Location on			Height above road surface measured from center of item on vehicle at curb weight
	Trucks	Trailers	Truck-tractors	
Intermediate side marker	On each side - 1 amber lamp located at or near midpoint between the front and rear side marker lamps (see 4.6.1.4).	Same as trucks		Not less than 15 inches, nor more than 60 inches.
Intermediate side reflex reflectors	On each side - 1 amber located at or near midpoint between the front and rear side reflex reflectors (see 4.6.1.4).	Same as trucks		Not less than 15 inches, nor more than 60 inches.
Light, warning, vehicular, rotating, DC	On top - near the centerline of the vehicle (see 4.6.1.5).		On top - near the centerline of the vehicle (see 4.6.1.5).	Not less than 24 inches, nor more than 60 inches.
Flashing warning lamps		On the rear - 1 red on each side of the vertical centerline at the same level and as far apart as practicable.		

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TABLE IV. Lighting equipment location for tactical military trucks, trailers and truck tractors of 80 or more inches overall width (see 3.21) - Continued.

Item	Location on			Height above road surface measured from center of item on vehicle at curb weight
	Trucks	Trailers	Truck-tractors	
Receptacle for CWL	Two waterproof receptacles located on the exterior of the back of the cab, one on each side of the centerline of the vehicle as far outboard as practicable, where the CWL may be plugged in.		Same as trucks.	

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TABLE V. Lighting equipment requirements and location for tracked vehicles.

Item	Quantity and color	Applicable SAE standards or military part numbers	Location	Recommended height above road surface measured from center of item on vehicle at curb weight
Headlamps	2 white	J580a, b MS18003-4811	At the same height, one on each side of the vertical centerline, at the same level, and as far apart as practicable.	Not less than 23 inches, nor more than 72 inches.
Taillamps	2 red	J585e 11614157 or 12287047	On the rear - 1 on each side of the vertical centerline, at the same level, and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.
Stoplamps	2 red	J586c 11614157 or 12287047	On the rear - 1 on each side of the vertical centerline, at the same level, and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.
Vehicular hazard warning lamps	2 red 2 amber		On the rear - 1 red on each side of the vertical centerline, at the same level, and as far apart as practicable. On the front - 1 amber on each side of the vertical centerline, at the same level and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.
Turn signal lamps	2 red 2 amber	J588e	At or near the front - 1 amber on each side of the vertical centerline, at the same level, and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.
		MS52125-2 11614157 or 12287047	On the rear - 1 red incorporated into a composite tail, stop and turn signal on each side of the vehicle centerline, at the same level and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.

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TABLE V. Lighting equipment requirements and location for tracked vehicles - Continued.

Item	Quantity and color	Applicable SAE standards or military part numbers	Location	Recommended height above road surface measured from center of item on vehicle at curb weight
Turn signal operating unit		J589 Apr 1964 11613632	Easy access of vehicle driver.	
Turn signal flasher	1	J590b 11613631		
Vehicular hazard warning signal operating unit	1	J910 Jan 1966 11613632		
Vehicular hazard warning signal flasher	1	J945 Feb 1966 11613631		
Clearance lamps on any vehicle whose headlights are more than 16 inches from the edge of the vehicle.	2 amber or white	J592e MS35423-1	On the front, one on each side of the vertical centerline not more than 16 inches from the edge of the vehicle.	Not less than 23 inches, nor more than 72 inches.
Reflex reflectors	2 red	J594f MS35387-1 11614174	On the rear, one red on each side of the vertical centerline, at the same level, as far apart as practical.	Not less than 15 inches, nor more than 60 inches.

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TABLE V. Lighting equipment requirements and location for tracked vehicles - Continued.

Item	Quantity and color	Applicable SAE standards or military part numbers	Location	Recommended height above road surface measured from center of item on vehicle at curb weight
Receptacle for CWL	1		On top of the vehicle or turret. Power receptacle must be weatherproof, separately fused and must not be hot when vehicle is in Black Out mode.	No requirement.
Flashing warning lamp	1 amber		Forward facing, separately switched on tank retrievers only.	Not less than 25 inches, nor more than 100 inches.

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TABLE VI. Military wheeled and tracked vehicle blackout lighting equipment requirements and location.

Item	Quantity and color		Military part numbers and photometric requirements	Location	Recommended height above road surface measured from center of item on vehicle at curb weight
	Tactical	Combat			
Blackout driving lamp 1/	1-white		11668932 (see 4.5.3)	The light source shall be mounted between the center and left-hand side of the vehicle, as far forward and as near the line of the driver's vision as practicable.	Approximately 42 inches.
		1-white or 1-white	MS51318-1 7972325 (composite) (see 4.5.3)		
Blackout front marker lamp 1/	2-white		MS52126-2 (composite) (see 4.5.2) 7972325 (see 4.5.2)	On the front - 1 on each side of the vertical centerline, at the same level, and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.
Blackout tail & stop	2-tail red 2-stop amber		MS52125-2 (see 4.5.4 & 4.5.5)	On the rear - 1 on each side of the vertical centerline, at the same level and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.
		1-tail red 1-stop amber	MS52125-2 (see 4.5.4 & 4.5.5)	On rear right side of the vertical centerline at the same level as the rear left side and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.
Blackout tail	1-red		MS52125-2 (see 4.5.4)	On rear left side of the vertical centerline at the same level as the rear right side, and as far apart as practicable.	Not less than 15 inches, nor more than 72 inches.

1/ Not required on trailers.

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TABLE VII. Military vehicle lighting equipment minimum candlepower requirements. 1/

Groups	Test Points Deg	Parking lamps	Group totals, CP								
			Tail lamps			Red stop and turn signal lamps			Yellow turn signal lamps		
Lighted compartments			One	Two	Three	One	Two	Three	One	Two	Three
1	20L-5U 20L-H 20L-5D 10L-10U 10L-10D	2.8	1.6	2.7	3.8	55	66	80	135	165	190
2	10U-V 5U-10L 5U-10R	2.4	2.1	3.6	5.5	85	100	115	210	251	290
3	10L-H 5L-5U 5L-5D	4.2	3.4	5.3	8.0	140	167	195	350	420	490
4	5U-V H-5L H-V H-5R 5D-V	16.8	9.6	16.5	24.0	380	449	520	950	1130	1295
5	5R-5U 5R-5D 10R-H	4.2	3.4	5.3	8.0	140	167	193	350	420	490
6	5D-10L 5D-10R 10D-V	2.4	2.1	3.6	5.5	85	100	115	210	251	290

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TABLE VII. Military vehicle lighting equipment minimum candlepower requirements - Continued. 1/

Groups	Test Points Deg	Parking lamps	Group totals, CP								
			Tail lamps			Red stop and turn signal lamps			Yellow turn signal lamps		
Lighted compartments			One	Two	Three	One	Two	Three	One	Two	Three
7	10R-10U 10R-10D 20R-5U 20R-H 20R-5D	2.8	1.6	2.7	3.8	55	66	80	135	165	190
Maximum - rear lamps only			15	20	25	300	360	420	900	900	900

1/ Grouped photometric minimum candlepower requirements for devices using one, two, or three separately lighted compartments, or for one, two, or three lamps used in a single design loction to perfom a single function.

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